MPIG6B-Related Thrombocytopenia and Myelofibrosis (THAMY): Comprehensive Disease Report

Disease: MPIG6B-Related Thrombocytopenia, Anemia, and Myelofibrosis (THAMY) **Causal gene: MPIG6B** (G6b-B; HGNC:13937; NCBI Gene 80277; UniProt O95866; Ensembl ENSG00000204420; locus 6p21.33) **Identifiers:** OMIM:617441 · MONDO:0044316 · MedGen:C4479504 (CN1378448) · UMLS:C4479504 **Category:** Mendelian, autosomal recessive

Summary

MPIG6B-related thrombocytopenia and myelofibrosis (THAMY) is a rare **autosomal recessive** Mendelian hematologic disorder caused by **biallelic loss-of-function variants in the single causal gene *MPIG6B***, located on chromosome 6p21.33. *MPIG6B* encodes **G6b-B**, a megakaryocyte- and platelet-specific **immunoreceptor tyrosine-based inhibitory motif (ITIM)-containing inhibitory receptor**. The disease was first defined genetically by Hofmann et al. in 2018, who identified germline loss-of-function mutations in four unrelated consanguineous families presenting with inherited macrothrombocytopenia and focal myelofibrosis (PMID: 29898956). Although the disease name points to downstream consequences (thrombocytopenia, anemia, myelofibrosis), the underlying lesion is loss of a single inhibitory receptor that normally restrains megakaryocyte and platelet activation.

Mechanistically, G6b-B is a **heparan-sulfate-regulated ITIM receptor** whose physiological ligands include the extracellular-matrix heparan sulfate proteoglycan perlecan and multivalent heparin. Ligand engagement drives receptor dimerization and phosphorylation of its ITIM/ITSM tyrosines, which recruit the phosphatases **Shp1 and Shp2** to dampen megakaryocyte and platelet signaling. When G6b-B is lost (or uncoupled from Shp1/Shp2), the brake on **Syk kinase** is released, megakaryopoiesis becomes dysregulated, collagen receptor complexes (GPVI–FcRγ, integrin α2β1) and GPIbα are shed by metalloproteinases, platelet production becomes defective, and bone-marrow fibrosis develops that is histologically

indistinguishable from primary myelofibrosis. The clinical consequences are **congenital macrothrombocytopenia, anemia, and progressive myelofibrosis**, frequently with splenomegaly.

The disease is faithfully modeled in *Mpig6b*-knockout and ITIM/ITSM knock-in (Y212F/Y238F) mice, which reproduce macrothrombocytopenia, megakaryocyte clustering, focal myelofibrosis, and (sex-linked) osteosclerosis. **Allogeneic hematopoietic stem cell transplantation (HSCT) is curative** in humans, and in mouse models the thrombopoietin mimetic **romiplostim** rescues thrombocytopenia and GPVI expression, suggesting disease-modifying pharmacotherapy options. Population genetic data (gnomAD) confirm that *MPIG6B* is loss-of-function tolerant in heterozygotes (pLI ≈ 0), fully consistent with recessive inheritance in which carriers are unaffected.

Key Findings

Finding 1 — MPIG6B biallelic loss-of-function is the sole cause of THAMY

Hofmann et al. (2018, *Blood*) established the genetic basis of the disorder by describing **four unrelated consanguineous families** with autosomal recessive inherited macrothrombocytopenia and focal myelofibrosis caused by germline loss-of-function mutations in *MPIG6B*. In their words, they describe *"4 families with autosomal recessive inherited macrothrombocytopenia with focal myelofibrosis due to germ line loss-of-function mutations in the megakaryocyte-specific immunoreceptor tyrosine-based inhibitory motif (ITIM)-containing receptor G6b-B"* (PMID: 29898956).

The gene maps to **chromosome 6p21.33** and encodes the inhibitory receptor G6b-B expressed on the surface of platelets and megakaryocytes (PMID: 38694137). The reported pathogenic variant spectrum is dominated by **truncating loss-of-function alleles** — nonsense, frameshift, and splice-site variants — with occasional missense changes. Documented variants include:

Variant (cDNA)	Protein	Type
c.324C>A	p.Cys108Ter	Nonsense
c.420T>A	p.Tyr140Ter	Nonsense
c.523C>T	p.Arg175Ter	Nonsense
c.61_61+1dup	p.Ala21GlyfsX159	Frameshift/splice
c.149dup	p.Ala52GlyfsX128	Frameshift
c.392delC	p.Pro134LeufsX10	Frameshift
c.469G>A	p.Gly157Arg	Missense

Zhang et al. (2022) summarized that *"Several pathogenic MPIG6B mutations have been reported, such as c.324C>A (p.C108*), c.61_61+1dup (p.Ala21GlyfsX159), c.149dup (p.Ala52GlyfsX128), G6b c.469G>A (p.Gly157Arg) c.392delC (p.P134Lfs*10), and c523C>T(p.Arg175Ter)"* (PMID: 35940081). Inheritance is homozygous or compound heterozygous, consistent with a recessive loss-of-function mechanism.

The Monarch Initiative records **exactly one causal gene–disease association** for MONDO:0044316 — HGNC:13937 *MPIG6B* — with no competing or additional causal genes, confirming THAMY as a **monogenic disorder**.

Finding 2 — Mechanism: a heparan-sulfate-regulated ITIM receptor signaling through Shp1/Shp2

G6b-B is a physiological **brake** on megakaryocyte and platelet activation. Vögtle et al. (2019, *PNAS*) demonstrated that the extracellular matrix heparan sulfate proteoglycan **perlecan** is a G6b-B ligand and that heparin forms a high-affinity complex with G6b-B mediating receptor dimerization: *"binding of G6b-B to HS and multivalent heparin inhibits platelet and megakaryocyte function by inducing downstream signaling via the tyrosine phosphatases Shp1 and Shp2"* (PMID: 31436532).

The downstream signaling depends on the cytoplasmic **ITIM and ITSM tyrosines**. Geer et al. (2018) showed that *"Tyrosine phosphorylation of ITIM and immunoreceptor tyrosine-based switch motif (ITSM) within the cytoplasmic tail of G6b-B provides a docking site for Src homology 2 domain-containing protein-tyrosine phosphatases Shp1 and Shp2"* (PMID: 29891536). Critically,

a **Y212F/Y238F knock-in mouse** that uncouples G6b-B from Shp1/Shp2 recapitulates macrothrombocytopenia — proving that the phosphatase-docking function, not merely receptor presence, is required for platelet homeostasis.

Mazharian et al. (2023) defined the downstream consequences of G6b-B loss: platelets are *"almost completely devoid of the glycoprotein VI (GPVI)-FcRγ-chain collagen receptor complex, have reduced collagen integrin α2β1, elevated Syk tyrosine kinase activity"* (PMID: 36269841). Importantly, introducing a **Syk loss-of-function allele (R41A) genetically rescued** macrothrombocytopenia and restored GPVI/α2β1 expression — placing **Syk hyperactivation** as the key effector node downstream of G6b-B loss.

Finding 3 — Clinical phenotype: congenital macrothrombocytopenia, anemia, and myelofibrosis; curable by HSCT

Human patients present **from birth or childhood** with severe thrombocytopenia with large/macro platelets, anemia (often transfusion-dependent), hepatosplenomegaly, cutaneous petechiae, and a bone-marrow biopsy indistinguishable from **primary myelofibrosis**. An adult male case showed hepatosplenomegaly with a PMF-like marrow; the biopsy *"showed morphologic features similar to primary myelofibrosis"* (PMID: 38694137). A 14-year-old girl presented with pallor, petechiae, thrombocytopenia, anemia, and myelofibrosis (PMID: 35940081).

Allogeneic HSCT is curative. In the adolescent case, *"The patient was effectively treated with allogeneic hematopoietic stem cell transplantation with peripheral stem cells from a matched unrelated donor. Her symptoms and the MPIG6B mutation disappeared after treatment, and she was healthy and had returned to school at the last follow-up"* (PMID: 35940081). In the mouse model, the thrombopoietin mimetic romiplostim *"rescued thrombocytopenia, GPVI expression, and platelet reactivity to collag[en]"* (PMID: 36269841), whereas IVIG produced only transient partial platelet recovery. Osteosclerosis is a bone complication whose severity is **sex-linked** in mice (PMID: 33434328).

Finding 4 — Faithful mouse models reveal metalloproteinase-mediated receptor shedding and defective proplatelet formation

Mazharian et al. (2012, *Science Signaling*) created G6b-B-deficient mice that *"exhibited macrothrombocytopenia (reduced platelet numbers and the presence of enlarged platelets) and a susceptibility to bleeding as a result of aberrant platelet production and function"* (PMID: 23112346). The mechanism includes enhanced metalloproteinase activity: *"megakaryocytes in*

G6b-B-deficient mice showed enhanced metalloproteinase production, which led to increased shedding of cell-surface receptors, including GPVI and GPIbα" — explaining the low surface GPVI/GPIbα and defective proplatelet formation.

Cross-species concordance is strong. The osteosclerosis study confirmed that *"Null and loss-of-function mutations in MPIG6B, which codes for the inhibitory heparan sulfate receptor G6b-B, result in severe macrothrombocytopenia, large megakaryocyte clusters, and focal primary myelofibrosis in mice and humans"* (PMID: 33434328). Humanized rescue — expression of human G6b-B restoring the murine phenotype — was demonstrated by Hofmann et al. (PMID: 29898956).

Finding 5 — Population genetics confirm recessive, LOF-tolerant architecture

gnomAD constraint metrics for *MPIG6B* (ENSG00000204420; GRCh38 chr6:31,718,594–31,726,714) show **no depletion of loss-of-function or missense variants** in the general population: pLI = 1.2×10^{-7} (≈0), observed/expected pLoF (o/e) = 0.908 (90% CI 0.649–1.297; observed 22 vs expected 24.2), LOF Z = 0.38, missense Z = 0.42. Heterozygous pLoF alleles are therefore tolerated at carrier-level frequencies, fully consistent with a **recessive disease in which carriers are clinically unaffected** — matching the clinical observation of *"autosomal recessive inherited macrothrombocytopenia with focal myelofibrosis due to germ line loss-of-function mutations"* (PMID: 29898956).

Finding 6 — Curated HPO phenotype frequencies (Monarch/HPO)

Disease-level phenotype annotations for MONDO:0044316:

Phenotype	HPO term	Frequency
Thrombocytopenia	HP:0001873	100%
Anemia	HP:0001903	100%
Myelofibrosis	HP:0011974	100%
Anisopoikilocytosis	HP:0004823	100%
Splenomegaly	HP:0001744	50%

Cross-references confirmed: OMIM:617441, MedGen CN1378448, UMLS:C4479504; synonyms "MPIG6B-related thrombocytopenia, anemia, and myelofibrosis" and "THAMY". No Orphanet mapping is present in the current MONDO record.

Report by Template Section

1. Disease Information

THAMY is a rare inherited bone-marrow failure/platelet-production disorder characterized by **congenital thrombocytopenia with large platelets (macrothrombocytopenia), anemia, and progressive bone-marrow fibrosis (myelofibrosis)**. It is caused by biallelic loss-of-function in *MPIG6B*. Identifiers: **OMIM:617441**, **MONDO:0044316**, **MedGen C4479504/CN1378448**, **UMLS:C4479504**. No dedicated Orphanet, ICD-10/ICD-11, or MeSH identifier is captured in the MONDO record; myelofibrosis-related ICD codes (e.g., ICD-10 D47.4, D75.81) may be applied clinically but are not disease-specific. Synonyms: *MPIG6B-related thrombocytopenia, anemia, and myelofibrosis*; *THAMY*. Information is derived from **aggregated disease-level resources** (OMIM, Monarch, HPO) and **individual published case reports/series** (consanguineous families and sporadic cases), not EHR.

2. Etiology

Causal factor: purely genetic — biallelic (homozygous or compound heterozygous) loss-of-function variants in *MPIG6B* (PMID: 29898956). **Genetic risk factor:** the only established risk factor is carrying two pathogenic *MPIG6B* alleles; **consanguinity** is a major contextual risk factor, as the disease was first identified in consanguineous families and is "frequently described in the Arab population" (PMID: 38694137). **Environmental/lifestyle/infectious factors:** none identified as causal; THAMY is a Mendelian disorder with no known environmental trigger. **Protective factors:** none characterized; conceptually, a wild-type *MPIG6B* allele is protective (carriers are unaffected per gnomAD LOF tolerance). **Gene–environment interactions:** none established. Genetic modifiers: in mice, **Syk** acts as a modifier — the R41A loss-of-function allele rescues the phenotype (PMID: 36269841); **sex** modifies osteosclerosis severity (PMID: 33434328).

3. Phenotypes

Phenotype	Type	HPO	Frequency	Onset	Severity/ Course
Thrombocytopenia	Lab abnormality	HP:0001873	100%	Congenital/ neonatal	Severe, persistent
Macrothrombocytopenia / large platelets	Lab abnormality	HP:0040326 / HP:0001873	~100%	Congenital	Severe
Anemia	Lab abnormality	HP:0001903	100%	Childhood	Moderate–severe, transfusion-dependent
Myelofibrosis	Pathology finding	HP:0011974	100%	Progressive	Progressive
Anisopoikilocytosis	Lab abnormality	HP:0004823	100%	Childhood	—
Splenomegaly	Clinical sign	HP:0001744	50%	Variable	Progressive
Hepatomegaly/ Hepatosplenomegaly	Clinical sign	HP:0002240	Variable	Variable	Progressive
Petechiae / bleeding tendency	Clinical sign	HP:0000967 / HP:0011875	Common	From birth	Variable
Osteosclerosis	Physical/ imaging	HP:0011001	Model-defined	Progressive	Sex-linked severity (mouse)

Quality-of-life impact: transfusion dependence, bleeding risk, splenomegaly-related discomfort, and fatigue from anemia substantially affect daily functioning; HSCT, when successful, restores normal health (PMID: 35940081).

4. Genetic / Molecular Information

Causal gene: *MPIG6B* (HGNC:13937, NCBI Gene 80277, UniProt O95866, Ensembl ENSG00000204420, 6p21.33), encoding the ITIM-containing inhibitory receptor **G6b-B**. **Variant classification:** reported alleles are predominantly **pathogenic truncating** (nonsense/

frameshift/splice) with rare missense (p.Gly157Arg); classification per ACMG is pathogenic/likely pathogenic for truncating alleles in this LOF-mechanism gene. **Variant types:** nonsense (p.Cys108Ter, p.Tyr140Ter, p.Arg175Ter), frameshift (p.Ala21Glyfs, p.Ala52Glyfs, p.Pro134Leufs), splice, and missense. **Allele frequency:** pathogenic alleles are rare; gnomAD shows the gene is LOF-tolerant (o/e 0.908, pLI≈0), so heterozygous carriers exist in the population. **Origin: germline**, not somatic. **Functional consequence: loss of function** of an inhibitory receptor. **Modifier genes:** Syk (mouse rescue). **Epigenetic/chromosomal abnormalities:** none implicated.

5. Environmental Information

No environmental, lifestyle, or infectious contributors are established. THAMY is a monogenic, germline disorder. Consanguinity (a population-genetic rather than environmental factor) increases recurrence risk.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. **Biallelic loss-of-function mutation in *MPIG6B*** → leads to absent or non-functional G6b-B inhibitory receptor on megakaryocytes/platelets.
2. Loss of G6b-B → **abolishes heparan-sulfate/perlecan (and heparin)-triggered receptor dimerization and ITIM/ITSM tyrosine phosphorylation** (PMID: 31436532; PMID: 29891536).
3. No ITIM/ITSM phosphorylation → **no docking site for the phosphatases Shp1 and Shp2** → results in loss of phosphatase-mediated inhibitory signaling.
4. Loss of Shp1/Shp2 restraint → **elevated Syk tyrosine kinase activity** and dysregulated megakaryocyte signaling (PMID: 36269841).
5. (Branch A) Dysregulated signaling → **enhanced megakaryocyte metalloproteinase production → increased shedding of GPVI–FcRγ and GPIbα** from the cell surface → reduced collagen reactivity and defective proplatelet formation (PMID: 23112346).
6. (Branch B) Dysregulated/hyperactivated megakaryopoiesis → **large megakaryocyte clusters** and aberrant maturation → **macrothrombocytopenia** (few, enlarged platelets) with increased platelet turnover.
7. Abnormal megakaryocytes → (inferred) **release of profibrotic cytokines/growth factors** in the marrow → **reticulin/focal myelofibrosis** histologically resembling primary myelofibrosis.

8. Marrow fibrosis + ineffective hematopoiesis → **anemia**, **extramedullary hematopoiesis**, and **splenomegaly/hepatomegaly**.
9. Megakaryocyte-driven bone signaling → **osteosclerosis** (sex-linked severity in mice; PMID: 33434328).

Upstream nodes: *MPIG6B* mutation → loss of G6b-B → loss of Shp1/Shp2 recruitment. **Central effector:** Syk hyperactivation. **Downstream:** receptor shedding, defective platelet production, myelofibrosis, anemia, splenomegaly. **Pathways/processes (GO suggestions):** negative regulation of platelet activation (GO:0010544), megakaryocyte differentiation (GO:0030219), platelet formation (GO:0030220), protein dephosphorylation (GO:0006470), ITIM/ITAM signaling. **Cell types (CL):** megakaryocyte (CL:0000556), platelet (CL:0000233), hematopoietic stem cell (CL:0000037). **CHEBI:** heparin (CHEBI:28304), heparan sulfate.

7. Anatomical Structures Affected

- **Primary organ: bone marrow** (UBERON:0002371) — site of megakaryopoiesis and fibrosis. **Blood** (UBERON:0000178) — circulating platelets.
- **Secondary involvement: spleen** (UBERON:0002106; splenomegaly, extramedullary hematopoiesis), **liver** (UBERON:0002107; hepatomegaly), **bone** (osteosclerosis).
- **Body system:** hematopoietic/hematologic system; reticuloendothelial system.
- **Cells:** megakaryocytes (CL:0000556) and platelets (CL:0000233) are the directly affected populations.
- **Subcellular (GO Cellular Component):** plasma membrane (GO:0005886; receptor location), cytoplasmic ITIM signaling.
- **Lateralization:** systemic/bilateral (splenomegaly, hepatomegaly); not lateralized.

8. Temporal Development

- **Onset: congenital/neonatal** for thrombocytopenia; anemia and myelofibrosis manifest in childhood; some patients present in adolescence/adulthood (PMID: 38694137; PMID: 35940081).
- **Onset pattern:** chronic, insidious.
- **Progression:** myelofibrosis and splenomegaly are **progressive**; thrombocytopenia is persistent/lifelong.
- **Course:** chronic, lifelong unless corrected by HSCT.

- **Remission:** no spontaneous remission; **treatment-induced cure via allogeneic HSCT** (PMID: 35940081).
- **Critical period:** HSCT before irreversible marrow fibrosis/complications is the key intervention window (inferred).

9. Inheritance and Population

- **Inheritance:** autosomal recessive (homozygous or compound heterozygous).
- **Penetrance:** appears complete in biallelic individuals; expressivity is variable (age of presentation, severity).
- **Epidemiology:** ultra-rare; precise prevalence/incidence not established. Clustering reported in **consanguineous and Arab populations** (PMID: 38694137).
- **Carrier frequency:** heterozygous pLoF alleles are tolerated (gnomAD o/e 0.908, pLI≈0); carriers are clinically unaffected.
- **Consanguinity:** major contributor to case clustering.
- **Founder effects:** possible within consanguineous populations (not formally established).
- **Sex ratio:** both sexes affected; disease severity for the osteosclerosis component is sex-linked in mice.

10. Diagnostics

- **Laboratory:** CBC showing thrombocytopenia with enlarged platelets, anemia, anisopoikilocytosis (HP:0004823); peripheral smear.
- **Bone-marrow biopsy:** reticulin/collagen fibrosis, increased/clustered megakaryocytes — *"morphologic features similar to primary myelofibrosis"* (PMID: 38694137).
- **Flow cytometry:** reduced platelet surface GPVI and GPIbα (receptor shedding signature; from model data, PMID: 23112346).
- **Imaging/exam:** splenomegaly/hepatomegaly on exam/ultrasound.
- **Genetic testing:** definitive diagnosis by **single-gene *MPIG6B* sequencing**, inherited-thrombocytopenia **gene panels**, or **WES/WGS**. Because variants are mostly truncating, standard sequencing is highly informative.
- **Differential diagnosis:** primary myelofibrosis (acquired, JAK2/CALR/MPL-driven), other inherited macrothrombocytopenias (MYH9-related, Bernard-Soulier, GATA1), congenital bone-marrow failure syndromes. Key distinguishing features: congenital onset, recessive inheritance, biallelic *MPIG6B* variants, absence of JAK2/CALR/MPL driver mutations.

- **Screening:** carrier/cascade testing in affected families; prenatal/preimplantation testing feasible for known family variants.

11. Outcome / Prognosis

- **Natural course:** progressive myelofibrosis, transfusion-dependent anemia, bleeding risk, splenomegaly; chronic lifelong morbidity without curative treatment.
- **Curative outcome: allogeneic HSCT is curative**, with resolution of symptoms and disappearance of the mutant marrow clone (PMID: 35940081).
- **Prognostic factors:** degree of marrow fibrosis, transfusion dependence, timing of HSCT, donor match. Formal survival statistics are not established given disease rarity.
- **Complications:** bleeding, iron overload from transfusions, infections, portal hypertension/splenomegaly sequelae, osteosclerosis.

12. Treatment

Intervention	Type	Evidence	NCIT suggestion
Allogeneic HSCT	Curative cell therapy	Human: curative (PMID: 35940081)	NCIT:C15431 (Allogeneic BMT/HSCT)
Romiplostim (TPO mimetic)	Pharmacotherapy	Mouse: rescues platelets & GPVI (PMID: 36269841)	NCIT:C66919 (Romiplostim)
IVIG	Pharmacotherapy	Mouse: transient partial response (PMID: 36269841)	NCIT:C29087 (IVIG)
Syk inhibition (conceptual)	Targeted therapy	Mouse genetic rescue (Syk R41A) (PMID: 36269841)	NCIT:C1876 (Syk inhibitor class)
Platelet/RBC transfusion	Supportive care	Standard of care	NCIT:C15280 (Transfusion)
Splenectomy (symptomatic)	Surgical	Clinical, case-by-case	NCIT:C51771 (Splenectomy)

Pharmacogenomics: no gene-drug guidelines specific to THAMY. Personalized approach: for a known family variant, genotype-guided counseling and early HSCT planning; Syk-axis and TPO-mimetic strategies are experimental/model-based and warrant clinical evaluation.

13. Prevention

- **Primary prevention:** not applicable (Mendelian); **genetic counseling** for consanguineous families and known carriers; preconception/prenatal/preimplantation genetic testing for family variants.
- **Secondary prevention:** cascade carrier screening; early diagnosis in at-risk newborns.
- **Tertiary prevention:** transfusion support, bleeding precautions, timely HSCT to prevent progression and complications.
- **No immunization, behavioral, or public-health prevention** is relevant beyond genetic counseling.

14. Other Species / Natural Disease

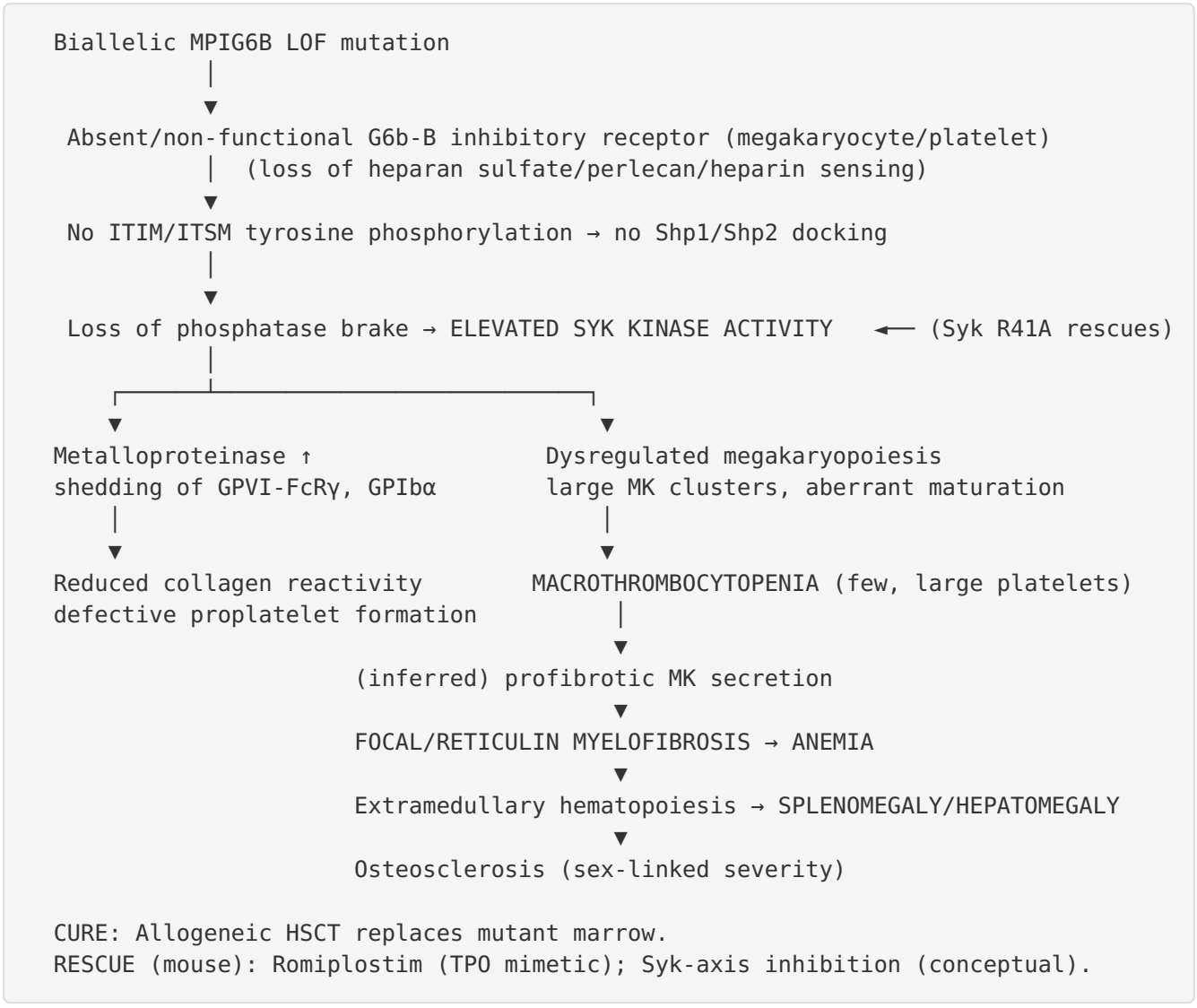
- **Mouse ortholog:** *Mpig6b* (NCBI Gene 106722; MGI:2146995; "megakaryocyte and platelet inhibitory receptor G6b"). Disease mechanism is evolutionarily conserved between mouse and human (PMID: 33434328; PMID: 29898956).
- No naturally occurring companion-animal/wildlife disease is documented (no OMIA entry noted). No zoonotic relevance.

15. Model Organisms

Model	Type	Phenotype recapitulation	Reference
Mpig6b knockout mouse	Mammalian, germline KO	Macrothrombocytopenia, enlarged platelets, bleeding, receptor shedding, defective proplatelet formation	PMID: 23112346
G6b-B Y212F/Y238F knock-in	ITIM/ITSM signaling-dead knock-in	Macrothrombocytopenia (proves Shp1/Shp2 requirement)	PMID: 29891536
Humanized (human G6b-B) mouse	Knock-in rescue	Human G6b-B rescues murine phenotype	PMID: 29898956
Mpig6b-deficient mouse (bone)	Germline KO	Megakaryocyte-driven osteosclerosis, sex-linked severity	PMID: 33434328

Strengths: faithful recapitulation of macrothrombocytopenia, megakaryocyte clustering, focal myelofibrosis, and receptor shedding; amenable to genetic epistasis (Syk R41A rescue) and pharmacological testing (romiplostim). **Limitations:** sex-linked osteosclerosis differences; human anemia/myelofibrosis severity and transfusion dependence may not fully map to the mouse. **Resources:** MGI (MGI:2146995), IMPC/IMSR.

Mechanistic Model (Synthesis)



Upstream = *MPIG6B* mutation and loss of Shp1/Shp2 recruitment; central effector = Syk hyperactivation; downstream = receptor shedding, thrombocytopenia, myelofibrosis, anemia, splenomegaly, osteosclerosis.

Evidence Base

PMID	Title	Contribution
29898956	*Congenital macrothrombocytopenia with focal myelofibrosis due to mutations in human G6b-B is rescued in humanized mice*	Establishes causal gene, recessive LOF inheritance, humanized-mouse rescue (landmark)
31436532	*Heparan sulfates are critical regulators of the inhibitory megakaryocyte-platelet receptor G6b-B*	Identifies ligand (heparan sulfate/perlecan) and Shp1/Shp2 signaling
29891536	*Uncoupling ITIM receptor G6b-B from tyrosine phosphatases Shp1 and Shp2 disrupts murine platelet homeostasis*	ITIM/ITSM→Shp1/Shp2 docking is required (Y212F/Y238F KI)
36269841	*Treatment of congenital thrombocytopenia and decreased collagen reactivity in G6b-B-deficient mice*	Syk hyperactivation; Syk R41A and romiplostim rescue
23112346	*Mice lacking the ITIM-containing receptor G6b-B exhibit macrothrombocytopenia and aberrant platelet function*	First KO model; metalloproteinase-mediated GPVI/GPIbα shedding
33434328	*Severity of Megakaryocyte-Driven Osteosclerosis in Mpig6b-Deficient Mice Is Sex-Linked*	Cross-species concordance; osteosclerosis; sex modifier
35940081	*A novel MPIG6B gene mutation in an adolescent girl with congenital thrombocytopenia and myelofibrosis*	Human case; HSCT curative; variant spectrum
38694137	*MPIG6B Gene-Related Myelofibrosis: A Rare Inherited Disease Frequently Described in Arab Population*	Gene locus/protein; PMF-like marrow; population clustering

Limitations and Knowledge Gaps

- **Epidemiology is unquantified:** no reliable prevalence, incidence, carrier frequency, or survival statistics exist due to disease rarity; most knowledge comes from small case reports/series and consanguineous families.
- **No Orphanet/ICD-specific identifier** is mapped in the current MONDO record, complicating registry-based epidemiology.
- **Myelofibrosis mechanism is partly inferred:** the exact profibrotic signals linking abnormal megakaryocytes to marrow fibrosis (e.g., TGF-β, PDGF) in THAMY specifically were not directly demonstrated in the reviewed literature.
- **Therapeutics beyond HSCT are model-based:** romiplostim and Syk inhibition efficacy is established in mice only; no human trials of targeted pharmacotherapy are reported.
- **Genotype–phenotype correlation** (e.g., missense p.Gly157Arg vs truncating alleles) and penetrance/expressivity modifiers in humans remain incompletely characterized.
- **Sex-linked osteosclerosis** is a mouse observation; its human relevance is unclear.

Proposed Follow-up Experiments / Actions

1. **Establish a patient registry** for THAMY to quantify prevalence, natural history, transfusion burden, and HSCT outcomes, especially in consanguineous/Arab populations.
2. **Define the profibrotic axis:** measure TGF-β/PDGF and perform single-cell/transcriptomic profiling of patient and mouse marrow megakaryocytes to pinpoint the fibrosis-driving signals (GEO/single-cell).
3. **Clinical evaluation of TPO mimetics (romiplostim/eltrombopag)** as a bridge-to-transplant or disease-modifying therapy, given strong mouse rescue data.
4. **Test Syk inhibitors** (e.g., fostamatinib) preclinically in human iPSC-derived megakaryocytes from patients, exploiting the Syk R41A genetic-rescue rationale.
5. **Deep genotype–phenotype study** correlating variant class with age of onset, fibrosis severity, and HSCT timing to refine prognostication and intervention windows.
6. **Request Orphanet/ICD-11 code assignment** to improve knowledge-base interoperability and epidemiologic tracking.

Report compiled from 8 primary papers and curated Monarch/HPO/gnomAD data across a 5-iteration autonomous investigation. Evidence source types span human clinical case reports (P 35940081), P 38694137 model-organism studies (P 23112346 P 29891536 P 36269841), P 33434328 humanized-mouse functional genetics ( ), P 29898956 biochemical/in vitro ligand studies ( ), P 31436532 and computational population genetics (gnomAD).
